



FINAL TEST SERIES for NEET-2024

MM : 720

Test-II

Time : 3 Hrs. 20 Mins.

Mock Test for NEET (Complete Syllabus of class XI & XII)

Instructions :

- There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
- Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
- Use blue/black ball point pen to dark appropriate circle.
-
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

PHYSICS

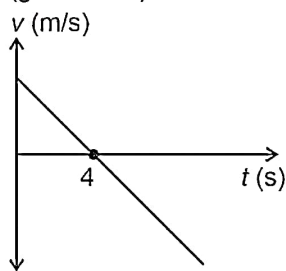
Choose the correct answer:

SECTION-A

1. In the expression for force (F), $F = \frac{at^2 + b}{c}$, if t is

time, then dimension of $\frac{a}{b}$ are

- (1) $[MLT^{-2}]$ (2) $[ML^0T^{-2}]$
 (3) $[ML^2T^{-2}]$ (4) $[M^0L^0T^{-2}]$
2. The given graph is for a particle projected upwards from a height of 40 m. The initial velocity of the particle is ($g = 10 \text{ m/s}^2$)



- (1) 20 m/s (2) 40 m/s
 (3) 80 m/s (4) 10 m/s

3. A bullet is fired horizontally from a 20 m high tower. If the speed of bullet is 100 m/s then how far from the foot of tower will it land?

(Take $g = 10 \text{ m/s}^2$)

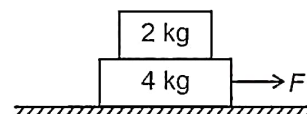
- (1) 20 m (2) 80 m
 (3) 160 m (4) 200 m

4. A particle moves in a circular path of radius 2 m such that its uniform angular speed is 4 rad/s. The ratio of its tangential acceleration and centripetal acceleration is

- (1) Zero (2) Infinite
 (3) 4 : 1 (4) 8 : 3

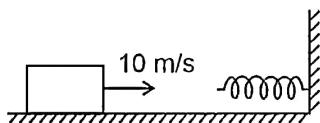
5. Two blocks of mass 4 kg and 2 kg kept one over the other on a smooth horizontal surface as shown below. If coefficient of friction between both blocks is 0.4 then the maximum force (F) for which they will move together will be ($g = 10 \text{ m/s}^2$)

- (1) 4 N
 (2) 8 N
 (3) 16 N
 (4) 24 N



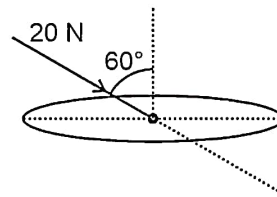
Test-11 (Code-A)

6. A stone of mass 200 g is tied to a 1 m long string and is whirled in a vertical circle such that it just completes the vertical circle, then ($g = 10 \text{ m/s}^2$)
- Tension at the highest point will be 2 N
 - Tension at the lowest point will be 12 N
 - Speed at the lowest point will be 10 m/s
 - Speed at the highest point will be zero
7. Moment of inertia of a ring of mass m and diameter d about an axis passing through its centre and perpendicular to its plane is
- md^2
 - $2md^2$
 - $\frac{md^2}{2}$
 - $\frac{md^2}{4}$
8. The correct relation between torque (τ) and angular momentum (L) is (Symbols have usual meanings)
- $\tau = \vec{r} \times \vec{L}$
 - $\vec{L} = \frac{d\vec{\tau}}{dt}$
 - $\Delta \vec{L} = \int \vec{\tau} \cdot dt$
 - $\vec{L} = \vec{r} \times \vec{\tau}$
9. A block of mass 250 g pushed towards a spring connected to a smooth platform with a speed of 10 m/s. If the spring constant of spring is 100 N/m, then the maximum compression in the spring will be

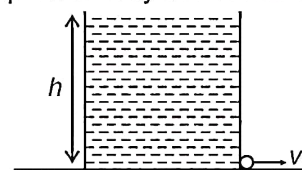


- 20 cm
 - 25 cm
 - 40 cm
 - 50 cm
10. P is a point at a distance r from the centre of a spherical shell of mass M and radius a , where $r < a$. The gravitational potential at P is
- $-\frac{GM}{r}$
 - $-\frac{GM}{a}$
 - $-\frac{GMr}{a^2}$
 - $-\frac{GM(a-r)}{a^2}$
11. The CGS unit of strain is
- dyne/cm²
 - dyne/cm
 - N/m²
 - No unit

12. A force of 20 N acts on a surface of area 40 cm² as shown below. The pressure exerted by this force is



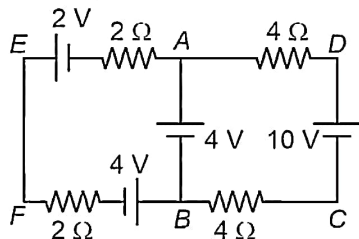
- 1 kPa
 - 2.5 kPa
 - 5 kPa
 - 10 kPa
13. From a container having liquid of density ' ρ ' upto height h , water comes out from an orifice made at its bottom (Area of cross-section = a). The initial thrust experienced by the container is



- $2gh\rho a$
 - $\frac{gh\rho a}{2}$
 - $2gha^2\rho$
 - $2gh\rho a^2$
14. One mole of an ideal diatomic gas undergoes an adiabatic expansion where its temperature changes from 500 K to 300 K, the work done by gas in this process is
- 250R
 - 100R
 - 500R
 - 750R
15. Motion of a particle executing SHM is expressed as $\frac{d^2x}{dt^2} + \frac{x}{16} = 0$ (in SI units)
- For an amplitude of 40 cm the maximum particle speed is
- 4 m/s
 - 0.4 m/s
 - 1 m/s
 - 0.1 m/s
16. A metallic wire of length 2 m is stretched by a tension corresponding to 10 kg weight. If mass of wire is 20 gram then the frequency of fundamental node formed will be
- 25 Hz
 - 500 Hz
 - 75 Hz
 - 100 Hz

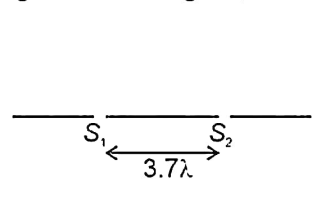
Space for Rough Work

17. Consider the following circuit and find the value of current flowing across AB.



- (1) 1.75 A (2) 2.5 A
(3) 3.25 A (4) 0.75 A
18. A particle of mass ' m ' and charge ' q ' enters a transverse uniform magnetic field B . In light of the given information choose the incorrect statement.
(1) Its velocity remains same
(2) Its kinetic energy remains same
(3) Work done by magnetic field on it is zero
(4) Its momentum keeps changing
19. The magnetic susceptibility of a magnetic substance is 1000, its relative permeability is
(1) 999 (2) 1001
(3) 99.9 (4) -1000
20. To convert a galvanometer into a ammeter one should connect a
(1) High resistance in series with galvanometer
(2) Low resistance in series with galvanometer
(3) High resistance in parallel with galvanometer
(4) Low resistance in parallel with galvanometer
21. The power factor of an LR circuit operating at AC source of frequency f is
(1) $\frac{R}{2\pi fL}$
(2) $\frac{R}{\sqrt{(2\pi fL)^2 + R^2}}$
(3) $\frac{R}{\sqrt{R + 2\pi fL}}$
(4) Zero

22. In a modified set-up of YDSE (as shown). The total number of minima, that can be formed on screen using a light of wavelength λ , are



- (1) 6 (2) 8
(3) 7 (4) 9
23. Consider the following statements and choose the correct option.
a. Mass defect in nucleus is proportional to its nuclear binding energy.
b. Binding energy per nucleon of a nucleus decides its stability.
(1) Only a is correct
(2) Only b is correct
(3) Both a and b are correct
(4) Both a and b are incorrect
24. For a light ray moving from denser to rarer medium a graph of angle of incidence vs angle of deviation is plotted as shown. Critical angle of the pair of media is
-
- (1) 30° (2) 40°
(3) 50° (4) 100°
25. The cut off wavelength for a continuous X-ray beam coming from an x-ray tube operating at 40 kV is
(1) 0.31 pm (2) 0.31 Å
(3) 0.62 pm (4) 0.62 Å

Space for Rough Work

Test-11 (Code-A)

26. If v is speed, r is radius and g is acceleration due to gravity, then which of the following is dimensionless?

- (1) $\frac{v^2 r}{g}$ (2) $\frac{v^2 g}{r}$
 (3) $\frac{v^2}{rg}$ (4) $v^2 rg$

27. The interference pattern is obtained with two coherent light sources of intensity ratio 4. In the

interference pattern, ratio $\frac{I_{\max} - I_{\min}}{I_{\max}}$ will be

- (1) 4 : 3 (2) 1 : 9
 (3) 8 : 9 (4) 5 : 4

28. Intensity of magnetisation of a magnetic specimen is basically the magnetic moment developed per unit volume. Its SI unit is

- (1) A m^2 (2) Tm A^{-1}
 (3) A m^{-1} (4) TA m^{-1}

29. In a plane electromagnetic wave the electric field oscillates sinusoidally at a frequency of 2×10^{10} Hz and amplitude of 48 V/m. The frequency of oscillation of magnetic field will be

- (1) 2×10^8 Hz (2) 16×10^{10} Hz
 (3) 2×10^{10} Hz (4) 16×10^8 Hz

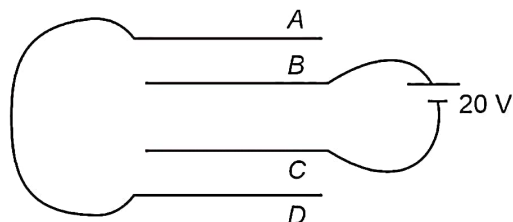
30. The capacity of a parallel plate capacitor depends on

- (1) The metal used to make the plate
 (2) The thickness of the plates
 (3) The potential applied across the plates
 (4) The separation between the plates

31. The magnifying power of an astronomical telescope for normal adjustment is 10 and the length of telescope is 110 cm. The magnifying power of the telescope when image is formed at the least distance of distinct vision is

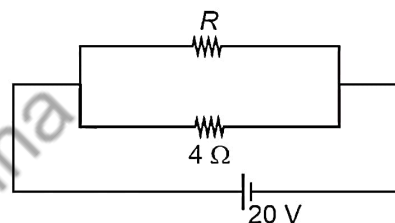
- (1) 14 (2) 48
 (3) 28 (4) 52

32. There are four identical metal plates each of area A . The plates are placed at equal distance ' d ' successively as shown in figure. If $\frac{\epsilon_0 A}{d} = 10 \mu\text{F}$ then equivalent capacity of the arrangement is



- (1) $15 \mu\text{F}$ (2) $7.5 \mu\text{F}$
 (3) $22.5 \mu\text{F}$ (4) $45 \mu\text{F}$

33. If in the circuit given below, total power dissipated is 300 W, then value of R is



- (1) 1Ω (2) 3Ω
 (3) 4Ω (4) 2Ω

34. Mutual inductance of a pair of coils is 0.8 H. If a current in the primary coil changes from 0.6 A to zero in 0.1 s then average induced emf in the secondary coil will be

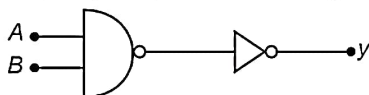
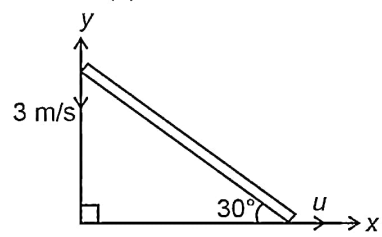
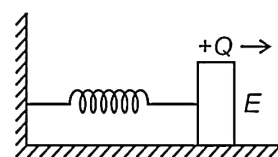
- (1) 4.8 V (2) 48 V
 (3) 24 V (4) 2.4 V

35. A ray of light strikes a glass plate at an angle of 30° with the glass surface. If the reflected and refracted rays are at right angles to each other then the refractive index of the glass is

- (1) 1.5 (2) 1.41
 (3) 1.7 (4) 1.33

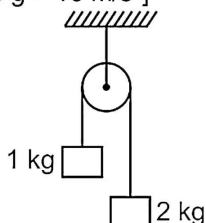
Space for Rough Work

SECTION-B

36. Which of the following quantities increase when wavelength of light incident on optical system is increased? (Consider only magnitudes)
- Power of a converging lens
 - Focal length of a converging lens
 - Focal length of a diverging lens
 - Both (2) and (3)
37. The work functions of sodium and copper are 2.0 eV and 4.0 eV respectively. Which metal will be suitable for a photoelectric cell using light of wavelength 4600 Å?
- Sodium
 - Copper
 - Both (1) and (2)
 - Neither (1) nor (2)
38. The energy of an electron of He^+ in the ground state is -54.4 eV, then the energy of an electron of Li^{++} in the first excited state will be
- 27.2 eV
 - -30.6 eV
 - -27.2 eV
 - -81.3 eV
39. Following diagram performs the logic function of
- 
- OR gate
 - NAND gate
 - NOR gate
 - AND gate
40. If $y_1 = \bar{A} + A$ and $y_2 = A \cdot \bar{A}$ then value of $y_1 + y_2$ in Boolean algebra is
- 1
 - Zero
 - A
 - $\bar{A}$
41. Nuclear reactions obey the law of conservation of
- Charge
 - Momentum
 - Mass-energy
 - All of these
42. Standing waves are produced in a 10 m long stretched string fixed at both ends. If the string vibrates in 5 segments and the wave velocity is 20 m/s, then the frequency of vibration is
- 6 Hz
 - 15 Hz
 - 5 Hz
 - 8 Hz
43. The rod as shown in the figure starts slipping at the instant when speed of upper end is 3 m/s and rod makes an angle of 30° with x-axis, speed of the lower end (u) of the rod at this instant is
- 
- 3 m/s
 - $\sqrt{3}$ m/s
 - 2 m/s
 - $\sqrt{2}$ m/s
44. The frequency of fundamental tone in a closed pipe is
- Same as the fundamental frequency of an open tube of same length
 - Half the frequency of fundamental tone of open tube of same length
 - Same as that of the first overtone of an open tube of same length
 - Double the frequency of fundamental tone of open tube of same length
45. A spring-block system undergoes a simple harmonic motion on smooth horizontal surface. The block is now given some positive charge Q and a uniform horizontal electric field to the right is switched on. As a result
- 
- The time period of oscillation will increase
 - The time period of oscillation will remain unaffected
 - The mean position of simple harmonic motion will shift to the right
 - Both (2) and (3)

Space for Rough Work

46. Two blocks of masses 1 kg and 2 kg are connected by a metal wire going over a smooth pulley as shown in figure. The breaking stress of the metal is $\frac{1}{3\pi} \times 10^9 \text{ N/m}^2$. What should be the minimum radius of the wire used, if it is not to break? [Take $g = 10 \text{ m/s}^2$]



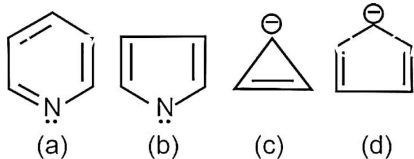
- (1) $2 \times 10^{-4} \text{ m}$ (2) $2 \times 10^{-8} \text{ m}$
 (3) $46 \times 10^{-8} \text{ m}$ (4) $23 \times 10^{-4} \text{ m}$
47. The magnetic susceptibility of a diamagnetic material at 227°C is -0.00002 . Then its magnetic susceptibility at 727°C will be
- (1) -0.00002 (2) -0.00004
 (3) -0.00001 (4) -0.00008

48. On the horizontal surface of truck a block of mass 5 kg is placed and coefficient of friction between the block and surface is $\mu = 0.4$. Maximum acceleration of truck for which the block will not move over the surface of truck is
- (1) 6 m s^{-2} (2) 5 m s^{-2}
 (3) 4 m s^{-2} (4) 8 m s^{-2}
49. The gravitational field at some point in space is $\vec{E}_g = 4\hat{i} + 5\hat{j} \text{ N/Kg}$. The force exerted on a 5 kg mass placed at a point in the space is approximately
- (1) 52 N (2) 45 N
 (3) 32 N (4) 125 N
50. 1 centipoise is equal to
- (1) $10 \text{ kg m}^{-1} \text{ s}^{-1}$
 (2) $1000 \text{ kg m}^{-1} \text{ s}^{-1}$
 (3) $0.01 \text{ kg m}^{-1} \text{ s}^{-1}$
 (4) $0.001 \text{ kg m}^{-1} \text{ s}^{-1}$

CHEMISTRY

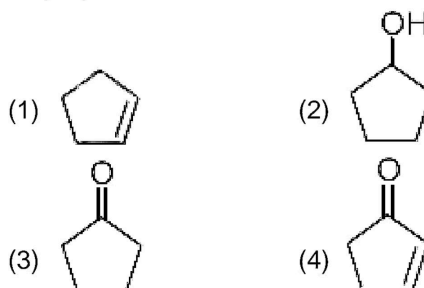
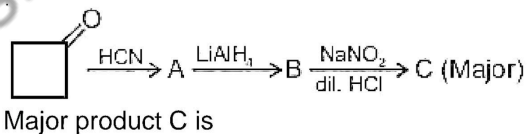
SECTION-A

51. If mass percentage of urea in aqueous solution is 15% then molality of urea in the solution is
- (1) 3.25 m (2) 6.15 m
 (3) 1.5 m (4) 2.94 m
52. Correct order of electron affinity of the given elements is
- (1) $F > Cl > O > S$ (2) $Cl > F > S > O$
 (3) $Cl > F > O > S$ (4) $F > Cl > S > O$
53. Which among the following compounds are aromatic in nature?



- (1) (a) and (b) only (2) (a), (b) and (c) only
 (3) (b), (c) and (d) only (4) (a), (b) and (d) only

54. Consider the following sequence



55. For which of the following reactions, $K_P = K_C$?
- (1) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{HCl}(\text{g})$
 (2) $\text{N}_2(\text{g}) + 2\text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
 (3) $2\text{NaHCO}_3(\text{g}) \rightleftharpoons \text{Na}_2\text{CO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{aq})$
 (4) $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

Space for Rough Work

56. A current of 2.0 ampere is passed for 5 hours through a molten salt of tin to deposit 22.2 g tin. The oxidation state of tin in its salt is

(At. wt. of tin = 119 g)

- (1) +1 (2) +4
(3) +2 (4) +3

57. pH of an aqueous solution of sodium acetate can be calculated as

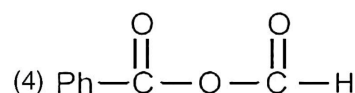
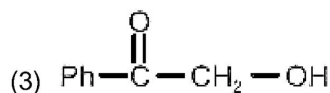
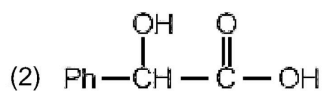
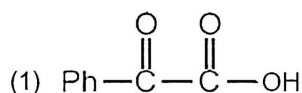
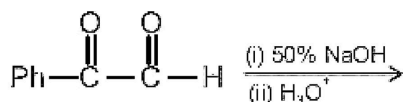
(1) $\text{pH} = 7 + \frac{1}{2}[\text{pK}_a + \log c]$

(2) $\text{pH} = 7 - \frac{1}{2}[\text{pK}_b + \log c]$

(3) $\text{pH} = 7 + \frac{1}{2}[\text{pK}_a + \text{pK}_b]$

(4) $\text{pH} = 7 - \frac{1}{2}[\text{pK}_a + \text{pK}_b]$

58. Major product of the given reaction is



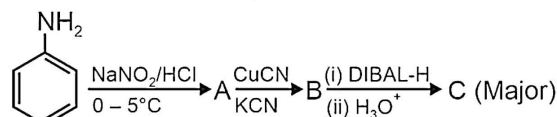
59. Which among the following is an incorrect statement?

- (1) Ionic radius of Co^{2+} is smaller than Cr^{2+}
(2) Manganese cannot show +5 oxidation state
(3) Co can form complex with CO
(4) Enthalpy of hydration of Fe^{2+} is higher than Mn^{2+}

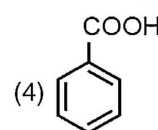
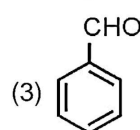
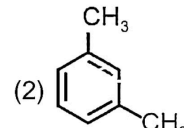
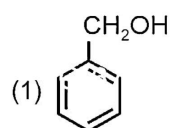
60. 99% of a first order reaction is completed in 32 minutes. Time required for 99.9% completion is

- (1) 50 minutes (2) 46 minutes
(3) 48 minutes (4) 64 minutes

61. Consider the following reaction.



Major product C is



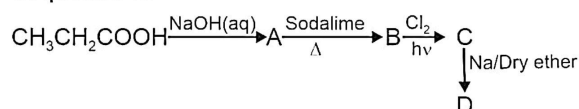
62. Most basic compound among the following is

- (1) $\text{Er}(\text{OH})_3$ (2) $\text{Gd}(\text{OH})_3$
(3) $\text{Sm}(\text{OH})_3$ (4) $\text{Ce}(\text{OH})_3$

63. Correct order of decreasing field strength of the given ligands is

- (1) $\bar{\text{CN}} > \text{CO} > \text{en}$ (2) $\text{CO} > \bar{\text{CN}} > \text{en}$
(3) $\text{en} > \text{CO} > \bar{\text{CN}}$ (4) $\text{CO} > \text{en} > \bar{\text{CN}}$

64. The major product (D) in the following reaction sequence is



- (1) Butane (2) But-2-ene
(3) Ethene (4) Ethyne

65. The extensive property among the following is

- (1) Heat capacity
(2) Boiling point
(3) Viscosity
(4) Density

Space for Rough Work

66. Consider the following statements.

Statement I : Isotonic solutions have same osmotic pressure at a given temperature.

Statement II : Reverse osmosis occurs when pressure smaller than osmotic pressure is applied to the solution side.

In the light of above statements, select the correct option.

- (1) Statement I is false but statement II is true
- (2) Both statement I and II are false
- (3) Both statement I and II are true
- (4) Statement I is true but statement II is false

67. The species having highest bond order is

- (1) O_2^-
- (2) N_2^+
- (3) NO^+
- (4) NO

68. The catalyst used in the Ostwald's process for the manufacture of HNO_3 is

- (1) Cr_2O_3
- (2) FeO
- (3) V_2O_5
- (4) Platinised asbestos

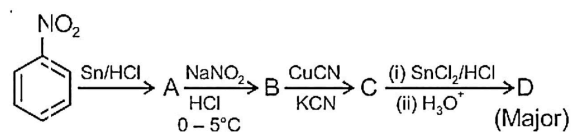
69. The metal carbonyl which reduced easily is

- (1) $V(CO)_6$
- (2) $Cr(CO)_6$
- (3) $Fe(CO)_5$
- (4) $Ni(CO)_4$

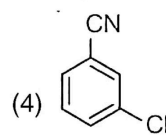
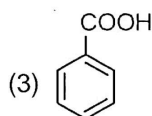
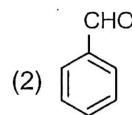
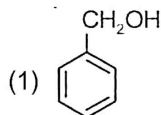
70. Which of the following relations is incorrect?

- (1) $\Delta S_{total} = 0$ [for adiabatic reversible process]
- (2) $\Delta G = \Delta G^\circ = 0$ (when $Q_c = K_c = 1$)
- (3) $w = 2.303 RT \log \frac{V_f}{V_i}$ (for isothermal expansion of an ideal gas)
- (4) $w_{adiabatic} = nC_v(T_f - T_i)$

71. Consider the following reaction.



Major product D is



72. Consider the following statements.

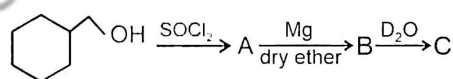
Statement-I: Methoxy group is ortho, para directing and deactivates the aromatic ring towards electrophilic substitution reactions.

Statement-II: The $-OH$ group attached to benzene ring activates the aromatic ring towards electrophilic substitution reactions.

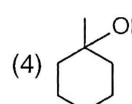
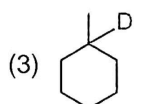
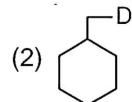
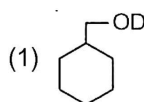
In light of above two statements, choose the correct option.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct and statement II is incorrect
- (4) Statement I is incorrect and statement II is correct

73. Consider the following reactions.



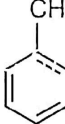
Product C is



74. Which among the following pair are globular proteins?

- (1) Keratin and insulin
- (2) Insulin and albumins
- (3) Keratin and myosin
- (4) Albumins and keratin

Space for Rough Work

75. Correct order of covalent character of given halides is
 (1) $\text{AlF}_3 > \text{AlCl}_3 > \text{AlBr}_3$ (2) $\text{AlCl}_3 > \text{AlBr}_3 > \text{AlF}_3$
 (3) $\text{AlF}_3 > \text{AlBr}_3 > \text{AlCl}_3$ (4) $\text{AlBr}_3 > \text{AlCl}_3 > \text{AlF}_3$
76. Mole of thiosulphate ions required to react completely with one mole of permanganate ion in faintly alkaline medium is
 (1) $\frac{2}{5}$ (2) $\frac{2}{3}$
 (3) $\frac{3}{8}$ (4) $\frac{3}{7}$
77. Cellulose is composed of
 (1) α -D-glucose units (2) β -D-glucose units
 (3) β -D-galactose units (4) β -D-fructose units
78. Consider the following statements.
Statement I : Boron is unable to form BF_6^{3-} ion.
Statement II : AlCl_3 achieves stability by forming a dimer.
 In the light of above two statements, choose the correct option.
 (1) Statement I is correct and statement II is incorrect
 (2) Both statement I and statement II are incorrect
 (3) Both statement I and statement II are correct
 (4) Statement I is incorrect and statement II is correct
79. Number of mole of CO_2 produced on heating 20 g of $\text{CaCO}_3(\text{s})$, given that CaCO_3 is only 75% pure is
 (1) $\frac{1}{20}$ (2) $\frac{3}{20}$
 (3) $\frac{3}{10}$ (4) $\frac{2}{23}$
80. Phosphonic acid has
 (1) Two P = O bonds (2) Two P – H bonds
 (3) Two P – OH bonds (4) One P – O – P bond
81. Hybridizations of atoms in the following compound in the sequence are
 $\text{CH}_2=\text{CH}-\dot{\text{N}}\text{H}-\text{CH}=\text{CH}-\dot{\text{N}}\text{H}_2$
 (1) (2) (3) (4) (5) (6)
 (1) $sp, sp^2, sp, sp^2, sp, sp^2$
 (2) $sp^2, sp^2, sp^3, sp^2, sp^2, sp^3$
 (3) $sp, sp^2, sp^3, sp, sp^2, sp^3$
 (4) $sp^2, sp^2, sp^2, sp^2, sp^2, sp^2$
82. Given below are four compounds.
 I. $\text{CH}_2=\text{CH}-\text{Br}$ II. $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Br}$
 III.  IV. $\text{CH}_3-\text{CH}_2-\text{Br}$
 Correct order of $\text{S}_{\text{N}}1$ reactivity of above compounds is
 (1) III > II > IV > I (2) II > III > IV > I
 (3) I > II > III > IV (4) III > IV > II > I
83. The group reagent for the test of alcohols is
 (1) Ceric ammonium nitrate
 (2) Baeyer's reagent
 (3) Ferric chloride
 (4) Tollens' reagent
84. DNA does not contain
 (1) Adenine (2) Guanine
 (3) Cytosine (4) Uracil
85. Number of oxygen atom(s) shared per SiO_4^{4-} unit in a pyrosilicate is
 (1) 1 (2) 2
 (3) 3 (4) 4
- SECTION-B**
86. The ratio of orbital angular momentum of electrons having $l = 2$ and $l = 3$ respectively will be
 (1) 2 : 1 (2) 1 : 2
 (3) $\sqrt{2} : 1$ (4) $1 : \sqrt{2}$
87. Maximum moles of Fe^{2+} that can be oxidised by 2 moles of dichromate ions in acidic medium is
 (1) 1 (2) 3
 (3) 6 (4) 12

Space for Rough Work

88. Which ion gives blood red colour with potassium thiocyanate?

- (1) Cd^{2+} (2) Cu^{2+}
(3) Fe^{3+} (4) Sn^{2+}

89. $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$ on heating loses 55.9% of its weight, value of x is

- (1) 2 (2) 5
(3) 6 (4) 10

90. IUPAC name of $\text{CH}_3\text{O}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$ is

- (1) 1-Methoxypropane (2) Methoxyisopropane
(3) Isopropoxymethane (4) 2-Methoxypropane

91. On monochlorination of 2-Methyl butane, the total number of optically active compounds formed is

- (1) 2 (2) 4
(3) 6 (4) 8

92. Following are the two statements one is labelled as assertion (A) and other is labelled as reason (R).

Assertion (A): Pressure does not have any significant effect on solubility of solids in liquids.

Reason (R): Solids and liquids are highly incompressible and remain unaffected by changes in pressure.

In the light of above statements, choose the correct option.

- (1) (A) is correct and (R) is incorrect
(2) (A) is incorrect and (R) is correct
(3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

93. Match the compounds given in list I with their shapes given in list II and identify the correct code.

	List I		List II
a.	PCl_5	(i)	Bent
b.	SO_2	(ii)	T-shaped
c.	SF_6	(iii)	Trigonal bipyramidal
d.	ClF_3	(iv)	Octahedral

- (1) a(i), b(iii), c(ii), d(iv) (2) a(iii), b(i), c(iv), d(ii)
(3) a(iv), b(iii), c(i), d(ii) (4) a(iii), b(iv), c(i), d(ii)

94. Match the column I of ions with column II containing their spin only magnetic moment and choose the correct option.

	Column I (Ions)		Column II (Magnetic moment in BM)
a.	Sc^{3+}	(i)	1.73
b.	Tl^{3+}	(ii)	0
c.	Tl^{2+}	(iii)	3.87
d.	V^{2+}	(iv)	2.84

- (1) a(ii), b(iii), c(i), d(iv) (2) a(i), b(ii), c(iii), d(iv)
(3) a(ii), b(i), c(iv), d(iii) (4) a(iv), b(iii), c(ii), d(i)

95. Following are the two statements labelled as assertion (A) and reason (R).

Assertion (A): Boiling point of pentane is higher than 2,2-Dimethylpropane.

Reason (R): Among isomeric alkanes, molecules becomes spherical with the increase in branching resulting in decrease in van der Waal forces of attraction.

In the light of above statements, choose the correct option.

- (1) (A) is correct and (R) is incorrect
(2) (A) is incorrect and (R) is correct
(3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

96. Number of sp^3 hybridised boron atom(s) in a borax molecule is

- (1) 1 (2) 2
(3) 3 (4) 4

97. Carbylamine test is not given by

- (1) $\text{CH}_3-\text{NH}-\text{CH}_3$
(2) CH_3NH_2
(3) $\text{C}_6\text{H}_5-\text{NH}_2$
(4) $\text{C}_2\text{H}_5\text{NH}_2$

Space for Rough Work

98. Consider the following statements.
- In a chemical reaction, if the structure of product is disordered than reactants, there will be increase in entropy.
 - For a given substance, the crystalline solid state is the state of highest entropy.
 - Entropy change is state function.

Which of the above statements are correct?

- (1) a and c only (2) a and b only
(3) b and c only (4) a, b and c only

99. Consider the given standard electrode potentials

$$E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.726 \text{ V}, E_{\text{Ag}^{+}/\text{Ag}}^{\circ} = +0.8 \text{ V}$$

$$E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = +0.345 \text{ V}, E_{\text{Fe}^{2+}/\text{Fe}}^{\circ} = -0.44 \text{ V}$$

Which among the following has highest oxidising power?

- (1) Ag^{+} (2) Fe^{2+}
(3) Zn^{2+} (4) Cu^{2+}

100. Consider the following statements.

- Many trivalent lanthanoid ions are coloured both in solid state and in aqueous solutions.
- Eu^{2+} is a strong reducing agent changing to common +3 state.
- Mischmetall is an alloy which consists of a 95% lanthanoid metal, 5% iron and traces of S, C, Ca and Al.

Which of the above statements are correct?

- (1) a and b only
(2) b and c only
(3) a and c only
(4) a, b and c

BOTANY

SECTION-A

101. Read the following statements and select the **correct** option.

Assertion (A): Ecosystems are not exempt from the second law of thermodynamics.

Reason (R): Flow of energy from sun to producers and then to consumer in an ecosystem is unidirectional.

- Both (A) and (R) are true and (R) is the correct explanation of (A)
- Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (A) is true but (R) is false
- (A) is false but (R) is true

102. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Axile placentation	(i)	Cabbage
b.	Basal placentation	(ii)	Okra
c.	Parietal placentation	(iii)	Primrose
d.	Free central placentation	(iv)	Wheat

- (1) a(ii), b(iv), c(i), d(iii) (2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(iv), c(i), d(ii) (4) a(iv), b(i), c(ii), d(iii)

103. All of the following features are associated with China rose, **except**

- Endospermic seeds
- Actinomorphic flowers
- Twisted aestivation in calyx
- Pentacarpellary ovary

104. Peripheral region of secondary xylem differs from central region, as the former is

- Involved in the conduction of water
- Resistant to the attacks of insects
- Dark brown in colour
- Hard, durable and resistant to attack of microbes

105. The endoplasmic reticulum

- Consist of parallelly stacked cristae
- Divides the intracellular space into two distinct compartments
- Lacks tubules and vesicles
- Can only be functional at acidic pH

Space for Rough Work

106. Dissolution of synaptonemal complex characterise the stage where
- (1) Nucleolus gets completely disappeared
 - (2) Chiasmata can be visualised
 - (3) Crossing over takes place
 - (4) Formation of tetrad takes place
107. Which of the following is **incorrect** w.r.t. xanthophylls?
- (1) It can absorb light and transfer energy to chlorophyll a
 - (2) It enables a wider range of wavelength of incoming light to be utilised
 - (3) It imparts yellow green colour in the chromatogram
 - (4) It protects chlorophyll a from photo-oxidation
108. The mobile carrier for transfer of electrons between complex III and IV is
- (1) A small lipid molecule located within the inner membrane
 - (2) Capable of oxidising NADPH
 - (3) Composed of FMN and two copper centres
 - (4) A small protein attached to the outer surface of the inner membrane
109. A sigmoid curve
- (1) Lacks stationary phase
 - (2) Is seen when nutrients are unlimited
 - (3) Can be represented by equation $L_t = L_0 + rt$
 - (4) Is a characteristic of living organism growing in a natural environment
110. β -thalassemia is similar to sickle cell anaemia in all of the following aspects, **except**
- (1) Being an autosomal recessive trait
 - (2) Being controlled by a single gene on chromosome 11
 - (3) Being a disorder associated with haemoglobin molecule
 - (4) Being a quantitative disorder
111. Read the following statements of Assertion (A) and Reason (R) and choose the **correct** option.
- Assertion (A):** Transcription and translation can be coupled in bacteria.
- Reason (R):** In bacteria, mRNA does not need processing and; transcription and translation take place in cytosol.
- (1) (A) is true but (R) is false
 - (2) (A) is false but (R) is true
 - (3) Both the statements (A) and (R) are true and (R) is the correct explanation of (A)
 - (4) Both the statements (A) and (R) are true but (R) is not the correct explanation of (A)
112. Choose the **incorrect** statement.
- (1) Exine of the pollen grains is made up of sporopollenin.
 - (2) Pollen grains of many species cause severe allergies.
 - (3) Pollen grains of rice and wheat lose viability within 30 minutes of their release
 - (4) In majority of angiosperms, pollens are shed at three-celled stage.
113. Identify the stages of cell cycle based on the events given below.
- A. Synthesis of nucleotides, ATPs, amino acids, RNAs
- B. Cell doubles its DNA amount
- C. Duplication of mitochondria and chloroplast
- | | A | B | C |
|-----|----------------------|----------------------|----------------------|
| (1) | G ₂ phase | G ₁ phase | M phase |
| (2) | S phase | G ₂ phase | G ₀ phase |
| (3) | G ₀ phase | S phase | G ₁ phase |
| (4) | G ₁ phase | S phase | G ₂ phase |
114. Eukaryotic flagella and cilia are similar to each other as both
- a. Have '9+2' organisation of microtubules
 - b. Are surrounded by membrane
 - c. Help in locomotion
 - d. Give cartwheel appearance with no microtubules in the centre.
- (1) Only a and b
 - (2) Only c
 - (3) Only a and d
 - (4) Only a, b and c

Space for Rough Work

115. Read the following statements and identify the **correct** ones.
- Intrafascicular cambium in dicot stems is primary in origin.
 - Heartwood is comprised of living cells with lignified walls.
 - The youngest layer of xylem lies just inside the vascular cambium.
- Only (a)
 - Only (b) and (c)
 - Only (a) and (c)
 - All (a), (b) and (c)
116. Read the following and choose the **incorrect** statement.
- Red algae show oogamous reproduction with complex post-fertilisation development.
 - Funaria* has multicellular and branched rhizoids.
 - Sequoia* is the smallest gymnosperm.
 - Cuscuta* is a parasitic flowering plant.
117. Insectivorous plants trap insects to meet their nitrogen needs, still they are placed in the kingdom Plantae because,
- They are multicellular
 - They are photosynthetic
 - They are eukaryotic
 - They are heterotrophic
- Only a and b
 - Only b
 - Only d
 - Only a, b and c
118. Viroids differ from viruses as the former
- Have genetic material
 - Have capsid
 - Are parasitic
 - Are devoid of protein coat
119. Select the **incorrectly** matched pair
- Microphylls – *Selaginella*
 - Macrophylls – Ferns
 - Leafy stage present – *Funaria*
 - Well-developed vascular tissues – *Marchantia*
120. How many of the given plants has/have zygomorphic flowers?
- Mustard, *Datura*, Chilli, Gulmohur, Bean
- Four
 - Three
 - Two
 - One
121. Read the following statements and select the **correct** option.
- Assertion (A):** Red-green colour blindness occurs in about 8% of males and only about 0.4% of females.
- Reason (R):** The genes that lead to red-green colour blindness are on the X chromosome
- Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true but (R) is not correct explanation of (A)
 - (A) is true but (R) is false
 - (A) is false but (R) is true
122. Select the **odd** one out w.r.t sex determination in honey bee.
- Males produce sperms by mitosis
 - Unfertilised egg develops as a drone
 - Males do not have father but can have sons
 - Males have half the number of chromosomes than that of a females
123. Which of the following mode of pollination is opted by seagrasses?
- Pollination occur in submerged condition by water
 - Male flowers are carried by water currents to female flowers at surface of water
 - Pollen grains in the air is trapped by stigma
 - Pollen grains are carried by rodents to the female flowers
124. DNA polymorphism can be defined as
- The mutations occurring only in the coding regions of the DNA
 - The mutations that have direct impact on the individual's reproductive ability
 - The allelic sequence variation in which more than one variant at a locus occurs in human population with a frequency greater than 0.01
 - The inheritable mutation that is observed in a population at minimal frequency

Space for Rough Work

125. Repetitive sequences in the human genome
- (1) Can never provide any evidence in tracing human history
 - (2) Shed light on chromosome structure, dynamics and evolution
 - (3) Make up less than 2% of the entire genome
 - (4) Code all types of proteins
126. Repressor gene of *lac* operon
- (1) Binds to the promoter region of the operon and prevents RNA polymerase from transcribing the operon
 - (2) Regulates the operon which is referred to as positive regulation
 - (3) Is activated by interaction with the inducer molecule
 - (4) Is synthesised constitutively
127. Statins produced by *Monascus purpureus*
- (1) Helps in removing clots from the blood vessels of patients who have undergone myocardial infarction
 - (2) Is used in detergent formulations
 - (3) Is used for clarifying bottled juices
 - (4) Acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol
128. Thermoregulation
- (1) Is energetically expensive for many organisms
 - (2) Is seen in green plants
 - (3) Can be easily performed by shrews inhabiting polar regions
 - (4) Is not a prominent property that is responsible for the success of mammals on earth
129. Select the **odd** one out w.r.t hotspots.
- (1) These are regions of accelerated habitat loss
 - (2) These are regions which show low degree of endemism
 - (3) These are regions which show high levels of species richness
 - (4) Currently the biodiversity hotspots in the world are 34.
130. In taxonomical hierarchy, lower the taxa
- (1) Greater is the difficulty of determining the relationship to other taxa at the same level
 - (2) Greater is the number of organisms within the group
 - (3) Greater is the rank in hierarchy
 - (4) More are the characteristics that the members within the taxon share
131. Protein rich layer named pellicle is present in
- (1) Euglenoids
 - (2) Diatoms
 - (3) Bacteria
 - (4) Sac fungi
132. Select the **odd** one w.r.t. organisms showing haplo-diplontic life cycle.
- (1) *Marchantia*
 - (2) *Spirogyra*
 - (3) *Ectocarpus*
 - (4) *Polysiphonia*
133. The supporting root coming out of the lower nodes of the stem is called
- (1) Tap root
 - (2) Stilt root
 - (3) Pneumatophore
 - (4) Primary root
134. Main functioning of the cytoskeleton in a cell includes all, **except**
- (1) Cell division
 - (2) Mechanical support
 - (3) Motility
 - (4) Photophosphorylation
135. Single membranous cell organelle is
- (1) Mitochondria
 - (2) Lysosome
 - (3) Chloroplast
 - (4) Ribosome
- SECTION-B**
136. Mitotic apparatus formation occurs during
- (1) Telophase
 - (2) Anaphase
 - (3) Metaphase
 - (4) Prophase
137. Karyokinesis is not followed by cytokinesis, it results into
- (1) Syncytium
 - (2) Multicelled condition
 - (3) Haploid cells
 - (4) Uninucleate cells
138. Select the **odd** one out w.r.t plant factors affecting photosynthesis
- (1) Orientation of leaves
 - (2) Temperature
 - (3) Amount of chlorophyll
 - (4) Size of leaves

Space for Rough Work

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138. Select the **odd** one out w.r.t plant factors affecting photosynthesis
- (1) Orientation of leaves
 - (2) Temperature
 - (3) Amount of chlorophyll
 - (4) Size of leaves

Space for Rough Work

139. The RQ value of tripalmitin is
 (1) 1.0 (2) 0.9
 (3) 0.7 (4) 1.4
140. All of the following are the reasons that causes seed dormancy, **except**
 (1) Presence of phenolic acids
 (2) Presence of immature embryos
 (3) Presence of nitrates
 (4) Presence of impermeable and hard seed coat
141. Chromosomal complement of a person affected from phenylketonuria is
 (1) 44 + XXY (2) 44 + XY or 44 + XX
 (3) 44 + XO (4) 44 + XXX
142. In ABO blood grouping system of human beings, maximum possible number of genotypes and phenotypes respectively are
 (1) 6 and 4 (2) 6 and 6
 (3) 4 and 6 (4) 4 and 4
143. During transcription, opening of DNA helix is facilitated by
 (1) DNA helicase (2) DNA polymerase
 (3) DNA ligase (4) RNA polymerase
144. Histones are basic proteins as they are rich in aminoacids
 (1) Lysine and Arginine
 (2) Methionine and Arginine
 (3) Methionine and Lysine
 (4) Lysine and Tryptophan
145. Primary sewage treatment process is
 (1) Physical process
 (2) Chemical process
 (3) Biochemical process
 (4) Physiological process
146. Population interaction in which one organism gets benefitted and other is harmed are
 (1) Parasitism and commensalism
 (2) Predation and mutualism
 (3) Predation and parasitism
 (4) Predation and commensalism
147. Major conduit of energy flow in terrestrial ecosystem is
 (1) Grazing food chain (2) Parasitic food chain
 (3) Detritus food chain (4) Auxillary food chain
148. Which of the following is **not** the alien species for India?
 (1) *Parthenium* (2) *Lantana*
 (3) *Eichhornia* (4) *Saccharum barberi*
149. Embryo sac of angiosperms
 (1) Has 3 synergids and 2 antipodal cells
 (2) Has three celled egg apparatus
 (3) Is a 8 celled and 7 nucleate structure
 (4) Has three antipodal cells at micropylar end
150. All of the following are components of periderm, **except**
 (1) Cork cambium
 (2) Vascular cambium
 (3) Cork
 (4) Secondary cortex

SECTION-A

151. Which of the following tissues has a limited role in absorption but provides protection against chemical and mechanical stresses?
 (1) Squamous epithelium
 (2) Compound epithelium
 (3) Cuboidal epithelium
 (4) Columnar epithelium
152. In humans, cells present in the interstitial spaces of testes secrete
 (1) Testosterone
 (2) Progesterone
 (3) FSH
 (4) LH

Space for Rough Work

153. Select the **incorrect** option w.r.t. the tissue that attaches skeletal muscles to bones.
- (1) Fibres and fibroblasts are compactly packed in this.
 - (2) Orientation of fibres show an irregular pattern.
 - (3) Collagen fibres are present in rows between many parallel bundles of fibres.
 - (4) This tissue is a dense regular connective tissue.
154. Which among the following elements is present in least amount in human body?
- (1) N (2) Na
 - (3) Si (4) O
155. All of the following statements are **correct** w.r.t. digestion in frogs, **except**
- (1) Pancreas produces pancreatic juice containing digestive enzymes.
 - (2) The wall of the stomach secretes HCl and gastric juices.
 - (3) Gall bladder synthesizes bile that emulsifies fat.
 - (4) Chyme is the partially digested food which passes from stomach to the first part of small intestine.
156. In humans, juxtaglomerular apparatus (JGA) is a special sensitive region formed by cellular modifications of
- (1) Proximal convoluted tubule and afferent arteriole
 - (2) Distal convoluted tubule and afferent arteriole
 - (3) Proximal convoluted tubule and efferent arteriole
 - (4) Distal convoluted tubule and efferent arteriole
157. Select the **odd** one w.r.t. the components of lymphatic system of frogs.
- (1) Lymph channel
 - (2) Lymph nodes
 - (3) Sinus venosus
 - (4) Lymph
158. Match column I with column II and select the **correct** option.
- | Column I | Column II |
|---------------------------|-------------|
| a. IUD | (i) Saheli |
| b. Barrier contraceptive | (ii) LNG-20 |
| c. Chemical contraceptive | (iii) Vault |
| d. Oral contraceptive | (iv) Foam |
- (1) a(ii), b(iii), c(iv), d(i) (2) a(iv), b(ii), c(iii), d(i)
 (3) a(ii), b(iv), c(i), d(iii) (4) a(iii), b(ii), c(iv), d(i)
159. Which of the following is a secondary metabolite that is not placed under the category of polymeric substances?
- (1) Gums (2) Lemon grass oil
 - (3) Cellulose (4) Rubber
160. Select the **correct** option w.r.t. the restriction sites present in *amp^R* gene of pBR322.
- (1) *Bam*H I and *Pst* I (2) *Pst* I and *Pvu* II
 - (3) *Pvu* I and *Pst* I (4) *Pvu* I and *Sal* I
161. In humans, under normal physiological conditions, which of the following factors does not favour the dissociation of oxygen from the oxyhaemoglobin?
- (1) High H⁺ concentration
 - (2) High pCO₂
 - (3) High temperature
 - (4) High pH
162. Which of the following events given below does not disrupts Hardy-Weinberg equilibrium?
- (1) Movement of individuals from one population to another
 - (2) Change in gene frequency by chance
 - (3) Absence of mutation
 - (4) Occurrence of natural selection
163. Which of the following statements is false w.r.t. human heart?
- (1) The walls of ventricles are thicker than that of atria.
 - (2) SA node is present in the lower left corner of the right atrium.
 - (3) The nodal musculature is autoexcitable.
 - (4) Heart normally beats 70-75 times in a minute.

Space for Rough Work

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Space for Rough Work

164. Choose the **odd** one w.r.t. chordates.
- (1) A post-anal tail is present.
 - (2) Pharynx is perforated by gill slits.
 - (3) Central nervous system is ventral, solid and double.
 - (4) Heart is ventral.
165. Complete the analogy by selecting the **correct** option w.r.t. adult humans.
120 mm Hg : Normal systolic pressure :: _____ : Normal diastolic pressure
- (1) 140 mm Hg (2) 60 mm Hg
 - (3) 80 mm Hg (4) 120 mm Hg
166. In *Periplaneta americana*, at the junction of A , a ring of B yellow coloured thin filaments is present that performs the function of excretion.
Select the **correct** option to fill in the blanks A and B.
- | A | B |
|------------------------|---------|
| (1) Foregut and midgut | 6-8 |
| (2) Foregut and midgut | 100-150 |
| (3) Midgut and hindgut | 6-8 |
| (4) Midgut and hindgut | 100-150 |
167. During the course of evolution, some of the land reptiles went back into water and evolved into fish-like reptiles, such as *Ichthyosaurs*. This event probably occurred around
- (1) 65 mya (2) 320 mya
 - (3) 200 mya (4) 350 mya
168. Read the statements given below carefully and select the option that represents **incorrect** statement.
- (1) *Ramapithecus* was more man-like while *Dryopithecus* was more ape-like.
 - (2) Two mya, *Australopithecines* probably lived in East Asian grasslands.
 - (3) The Neanderthal man lived between 1,00,000-40,000 years back.
 - (4) Agriculture came around 10,000 years back.
169. Select the **incorrect** statement w.r.t. respiratory diffusion membrane in humans.
- (1) The diffusion membrane is made up of three cellular layers.
 - (2) Total thickness of diffusion membrane is much less than a millimeter.
 - (3) Thin squamous epithelium of alveoli is a cellular layer.
 - (4) The basement substance is present in between the thin squamous epithelium of alveoli and the endothelium of alveolar capillaries.
170. Which among the following is an amino-acid derivative hormone?
- (1) GnRH (2) TSH
 - (3) Epinephrine (4) Thyrocalcitonin
171. In human males, the region outside the seminiferous tubules contains all of the following, **except**
- (1) Interstitial cells
 - (2) Immunocompetent cells
 - (3) Blood vessels
 - (4) Sertoli cells
172. Drug cocaine is obtained from the extracts of
- (1) *Cannabis sativa*
 - (2) *Papaver somniferum*
 - (3) *Erythroxylum coca*
 - (4) *Atropa belladonna*
173. Read the statements given below and select the **correct** option.
- Statement A:** Joints are essential for all types of movements, except locomotory movements
- Statement B:** Force generated by the muscles is used to carry out movements through joints.
- (1) Only statement A is correct
 - (2) Both statements A and B are incorrect
 - (3) Both statements A and B are correct
 - (4) Only statement B is correct
174. The enzyme that is generally used in extension step of PCR is isolated from
- (1) *Escherichia coli*
 - (2) *Thermus aquaticus*
 - (3) *Salmonella typhimurium*
 - (4) *Agrobacterium tumefaciens*

Space for Rough Work

175. Select the **incorrect** match w.r.t. disorders of muscular and skeletal system.

- (1) Gout – Accumulation of uric acid crystals in joints
 (2) Arthritis – Inflammation of joints
 (3) Osteoporosis – Increased estrogen levels
 (4) Tetany – Due to low Ca^{++} in body fluid

176. Match the zoological name of organisms listed in column I with their common name in column II.

	Column I		Column II
a.	<i>Chelone</i>	(i)	Garden lizard
b.	<i>Calotes</i>	(ii)	Magur
c.	<i>Clarias</i>	(iii)	Pigeon
d.	<i>Columba</i>	(iv)	Turtle

Choose the **correct** option.

- (1) a(iv), b(iii), c(ii), d(i) (2) a(iii), b(ii), c(iv), d(i)
 (3) a(iv), b(i), c(ii), d(iii) (4) a(i), b(iv), c(iii), d(ii)

177. In humans, the limbic system is a complex structure formed by all of the following, **except**

- (1) Hippocampus
 (2) The inner parts of cerebellar hemispheres
 (3) Amygdala
 (4) The inner parts of cerebral hemispheres

178. Choose the **incorrect** statement w.r.t. electrical synapses.

- (1) Transmission of impulse across them is similar to impulse conduction along a single axon.
 (2) The membrane of pre- and post-synaptic neurons are in close proximity.
 (3) They are abundant in our body.
 (4) Impulse transmission across an electrical synapse is always faster than that across a chemical synapse.

179. Select the **correct** option w.r.t. human neural system

- (1) Visceral nervous system comprises the whole complex of nerves, fibres, ganglia and plexuses.
 (2) The somatic neural system relays impulses from CNS to smooth muscles of the body.
 (3) The autonomic neural system transmits impulses from CNS to skeletal muscles.
 (4) The peripheral nervous system comprises of all the nerves of the body associated with the brain only.

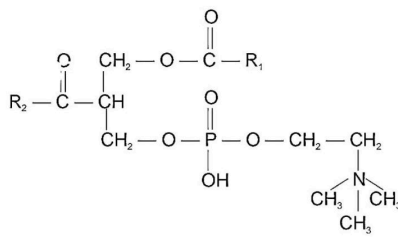
180. In humans, saddle joint is present between

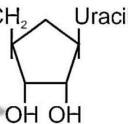
- (1) Humerus and pectoral girdle
 (2) Atlas and axis
 (3) Carpal and metacarpal of thumb
 (4) Carpals

181. The structure **not** associated with the brain stem of humans is

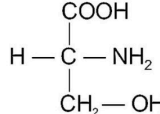
- (1) Pons (2) Cerebellum
 (3) Midbrain (4) Medulla oblongata

182. Which of the following is **incorrect** match?

- (1)  : Lecithin

- (2)  : Uridylic acid

- (3) $\text{CH}_3-(\text{CH}_2)_{14}-\text{COOH}$: Palmitic acid

- (4)  : Serine

183. Select the **incorrect** statement w.r.t. hormones secreted from GIT.

- (1) They are chemically peptides.
 (2) GIP inhibits gastric secretion and motility.
 (3) Secretin stimulates the secretion of pancreatic enzymes and bile juice.
 (4) The secretion of HCl and pepsinogen is stimulated by gastrin.

184. Volume of air that remains in lungs after a normal expiration is termed as

- (1) Residual volume
 (2) Tidal volume
 (3) Functional residual capacity
 (4) Expiratory capacity

Space for Rough Work

185. Select the **incorrect** option w.r.t. androgens.

- (1) These hormones stimulate muscular growth.
- (2) They play a major inhibitory role in the process of spermatogenesis.
- (3) They act on CNS and influence the male sexual behaviour.
- (4) These hormones produce anabolic effects on protein and carbohydrate metabolism.

SECTION-B

186. Which of the following can be a method to cure adenosine deaminase (ADA) deficiency permanently?

- (1) Periodic infusion of genetically engineered lymphocytes having functional ADA cDNA
- (2) Bone marrow transplantation
- (3) Enzyme replacement therapy
- (4) Introduction of ADA gene into cells at early embryonic stages

187. Which of the following facts/features is **correct** w.r.t. white muscle fibres?

- (1) Myoglobin is absent
- (2) Plenty of mitochondria is present
- (3) High amount of sarcoplasmic reticulum is present
- (4) They depend on aerobic process for energy

188. **Assertion (A):** Testes in human males are extra abdominal.

Reason (R): Lower temperature in scrotal sac than that of internal body temperature is necessary for spermatogenesis.

In the light of above statements, select the **correct** option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true, (R) is false
- (4) Both (A) and (R) are false

189. Choose an **incorrect** statement w.r.t. mammary glands.

- (1) A functional mammary gland is the characteristic of all members of class Mammalia including humans.
- (2) The mammary glands contain glandular tissue and variable amount of fat.
- (3) The glandular tissue of each breast is divided into 15-20 mammary lobes.
- (4) Each mammary lobe contains cluster of cells called alveoli.

190. Leydig cells are acted upon by the hormone secreted by _____ to stimulate the secretion of androgens.

Select the **correct** option to fill in the blank.

- (1) Pars nervosa
- (2) Neurohypophysis
- (3) Pars intermedia
- (4) Pars distalis

191. According to MTP (Amendment) act 2017, the opinion of how many registered medical practitioner(s) is/are required for MTP, if the pregnancy has lasted for seven weeks?

- (1) Two
- (2) Three
- (3) Zero
- (4) One

192. Which one of the following **correctly** explains the function of a specific part of human nephron?

- (1) Distal convoluted tubule – Reabsorption of K^+ into the renal tubules
- (2) Afferent arteriole – Carries blood away from the glomerulus and towards renal vein
- (3) Podocytes – Create slit pores for the filtration of blood into the Bowman's capsule
- (4) Henle's loop – Maximum reabsorption of major substances from the glomerular filtrate

193. All the following methods of ART involve *in-vitro* fertilisation, **except**

- (1) ICSI
- (2) ZIFT
- (3) GIFT
- (4) IUT

194. Choose the structure **not** found in a male cockroach.

- (1) Labium
- (2) Anal cerci
- (3) Anal style
- (4) Spermatheca

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195. Observe the figure given below and select the **correct** option w.r.t. it.



- (1) Drugs obtained from this plant are opioids
 (2) Their receptors are present principally in the brain
 (3) Usually taken by snorting and injection
 (4) The plant is native to South America
196. Read the following statements A and B w.r.t. RCH.
Statement-A: It creates awareness among people about various reproduction related aspects.
Statement-B: It provides facilities and support for building up a reproductively healthy society.
 Choose the **correct** option.
 (1) Both statements A and B are correct
 (2) Both statements A and B are incorrect
 (3) Only statement A is correct
 (4) Only statement B is correct
197. Select the **incorrect** statement among the following w.r.t. cancer.
 (1) Cancer cells do not show the property of contact inhibition.
 (2) Cancer causing viruses are called viral oncogenes.
 (3) Metastasis is the most feared property of malignant tumors.
 (4) Transformation of normal cells into cancerous cells may be induced by physical, chemical or biological agents.

198. Choose the **incorrect** statement w.r.t. blood vessels.

- (1) Arteries have no valves.
 (2) Veins are collapsible because they have thin walls.
 (3) Tunica intima of arteries has a thin muscular membrane with few elongated endothelial cells.
 (4) Tunica media is composed of smooth muscles and elastic fibres.

199. All of the following statements are incorrect w.r.t common diseases in humans, **except**.

- (1) Common cold is a bacterial disease caused by *Haemophilus influenzae*.
 (2) *Epidermophyton* is a protozoan which causes amoebiasis.
 (3) Bacteria like *Streptococcus pneumoniae* is responsible for the disease pneumonia.
 (4) *Salmonella typhi* is a virus which causes typhoid fever

200. Read the statements carefully and select the **correct** option w.r.t. humans.

Statement A: The descending limb of loop of Henle is permeable to water but almost impermeable to electrolytes.

Statement B: PCT helps to maintain pH of blood by secretion of HCO_3^- .

- (1) Both the statements A and B are correct
 (2) Statement A is correct but B is incorrect
 (3) Statement B is correct but A is incorrect
 (4) Both the statements A and B are incorrect



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